Is the polarization index a valid measure of loyalty for evaluating changes over time?

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Abstract

Purpose – This paper aims to argue that the polarization index (φ) represents a valid loyalty measure for evaluating changes over time.

Design/methodology/approach – The brand performance measures (BPM) are a valid and useful tool for marketing managers in measuring the loyalty consumers attach, in a single time period, to a product or brand. However, the BPM reflect other attributes and not only loyalty. Over time, what might appear to be a change in loyalty may actually be a change in market size or market share. The polarization index (φ) is not biased in this manner and is more appropriate for evaluating changes over time. The study compares the results obtained with three well known BPM utilised for the analysis of loyalty – the purchase frequency, the share of category requirements and the repeat rate – with those obtained with the φ on the purchases of wine made by Italian consumers in the retail sector over two three-year periods (2003-2005 and 2006-2008).

Findings – The study shows that the BPM are a fundamental source of information on the loyalty consumers attach to brands and products at one point in time. However, their strong relationship with market share risks providing results that do not reflect actual trends in loyalty. By comparison, φ provides a valid and useful analysis of the ways in which loyalty evolves over time.

Originality/value – Although several researchers have studied the uses of φ on one-year and three-year periods, none observed how the index offers more valid results than the BPM over time. The paper shows that marketing managers should always compare the results obtained with the BPM with those derived from the φ before drawing conclusions on the real loyalty trends of their products and brands.

Keywords Brands, Brand loyalty, Wines, Customer loyalty, Consumer behaviour

Paper type Research paper

An executive summary for managers and executive readers can be found at the end of this article.

1. Introduction

Marketing managers often analyse the brand performance measures (BPM) in order to understand trends for their products and brands over time (Sudman and Wansink, 2002; Ehrenberg *et al.*, 2004). Some of these measures appear to be of particular interest, as they provide information on consumers' behavioural loyalty levels: the purchase frequency (PF), the share of category requirements (SCR) and the repeat purchase rate (ρ) (Ehrenberg *et al.*, 2004; Jarvis *et al.*, 2007a; Singh *et al.*, 2009). Models have been developed for the relationships between the BPM, finding that products or brands with higher market share (MS) tend also to have a higher PF, a higher SCR and a higher ρ (Goodhardt *et al.*, 1984; Uncles *et al.*, 1995; Ehrenberg *et al.*, 2004). Said differently, these BPM are all functions of the MS. However, this association does not always hold true (Fader and Schmittlein, 1993; Kahn *et al.*, 1988; Jarvis and Goodman, 2005; Jarvis *et al.*, 2007a, b); therefore, the evaluation of the loyalty levels based on these measures may not correspond to reality (Jarvis *et al.*, 2007a).

In order to resolve this issue, researchers developed an index able to provide a measure of loyalty independent of the MS: the polarization index (φ) (Sabavala and Morrison, 1977; Rungie, 2000). This measure found a fruitful field of application in the study of consumer behaviour towards wine, with articles published in different countries such as Australia (Jarvis and Goodman, 2005; Jarvis *et al.*, 2006, 2007a, b), Greece (Krystallis and Chrysochou, 2009) and Italy (Casini *et al.*, 2008, 2009).

However, all of these studies applied the φ on either a one-year or a three-year period. None of them observed whether the index offers more valid results with respect to changes in the BPM over time, although it was pointed out that if the analysis of loyalty is important, it is even more important to understand how loyalty evolves over time (Bloemer and Kasper, 1995; Olsen, 2002; Jarvis *et al.*, 2003; Ehrenberg *et al.*, 2004).

By comparing the results obtained using the BPM with those generated by the φ on the purchases of wine made by Italian consumers in the retail sector over two consecutive three-year periods (2003-2005 and 2006-2008), this study pursues two specific objectives. On a theoretical side, the research aims to verify that the φ represents a more valid measure of loyalty than BPM over time. This objective also has managerial implications. The BPM are a valid and useful tool for marketing managers, but if they provide biased estimates of the changes in the loyalty consumers attach to a certain product or brand, then analysts should examine the evolution of φ values over time instead.

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The Italian retail wine sector was chosen for two main reasons. Most of the research published thus far on the φ and its relationship with BPM are on wine sold in the retail channels, thus providing more references about how to understand and interpret the outcomes of the φ. However, this sector is facing a particularly critical situation. According to Euromonitor International (2008), the consumption of wine is declining in Italy, France, Spain and Argentina, amongst other countries. Further, Switzerland and Spain (ranked eighth and tenth largest by value, respectively, in 2002) dropped out of the top ten wine drinking countries and were replaced by Canada and Russia. Australia has been reducing the grape surplus, reducing bulk export shipments and driving up prices. India, China and the UK are emerging as the new wine producing countries. In this context, the ability to understand the elements driving consumers' loyalty over time appears to be more than a simple opportunity, and not only for the Italian market.

This article is organised as follows. Following this introduction, a literature review will be presented on the reasons the analysis of loyalty and its evolution over time is such an important topic for marketing managers. The BPM will be briefly described. The reasons why empirical patterns do not always reflect real changes in loyalty will be explained. The advantages and the main results obtained with the φ will be discussed. Then, it will be demonstrated how to calculate both the BPM and φ. This is followed by the presentation of the results and discussion. Finally, conclusions and some suggestions for future research are presented.

2. Literature review

Loyalty represents a fundamental concept in marketing (Bennet *et al.*, 2007) and a key issue in business (Smith *et al.*, 2004). Retaining customers is beneficial for firms (Gwinner *et al.*, 1998; Brown, 2004; McMullan, 2005; Plesko, 2006; Chao, 2008). Bennet (2002) summarises the advantages of having loyal customers for a firm in the following points:

- they tend to spend more than non-loyal customers and, thanks to positive word-of-mouth, they may act as advocates for firms;
- it costs less to maintain current customers than to acquire new ones;
- loyal customers are more likely to purchase than newcomers;
- they are less deal prone;
- they place more frequent and similar orders, hence they cost less to serve;
- customer and employee loyalty are highly correlated;
- knowing the preferences of loyal customers make marketing activities more efficient; and
- customers are more prone to give a second chance to firms in case of a failure.

Moreover, being loyal to a firm represents an advantage for customers, as they can minimise the risk associated with a new purchase (Sweeney and Swait, 2008), they may receive a higher level of customisation (Gremler *et al.*, 1997) and they may gain a higher feeling of familiarity with the product or service provider (Reinartz and Kumar, 2002).

However, loyalty is not stable; it evolves over time (Bloemer and Kasper, 1995; Olsen, 2002). Mittal *et al.* (2001) observed a temporal change in an attribute's importance by analysing overall satisfaction towards a car company two months after the purchase and 24 months later. Johnson *et al.* (2006) confirmed this, by studying the cellular phone market in Germany in 1996, 1998 and 2000. Löfgren *et al.* (2008) stated that benefits and attributes have different roles in affecting customer satisfaction and loyalty, in respect to the different moments of the consumption cycle of three packaged goods: ground coffee, frozen lasagne and orange juice.

Three BPM, in particular, were found able to provide information on the degree of loyalty consumers devote to specific products or brands: the PF[1], the SCR[2] and the ρ[3] (Ehrenberg *et al.*, 2004; Jarvis *et al.*, 2007a; Singh *et al.*, 2009). However, these measures are not exempt from limitations. They are often expensive and time consuming to collect (Uncles and Lee, 2006), they tend to overlap on the kind of information they provide to analysts (Lehman *et al.*, 2008), and it is not often clear what should be the standard time period over which these measure should be collected (Jarvis *et al.*, 2003). Further, the interpretation of the results is not often straightforward, as, for example, an SCR of 50 per cent could have been the result of 50 per cent of the buyers purchasing it exclusively and 50 per cent purchasing it occasionally, or 100 per cent of buyers purchasing it about one half of the times (Jarvis *et al.*, 2003). Moreover, although well known and established patterns between these BPM have been identified (Goodhardt *et al.*, 1984; Ehrenberg *et al.*, 2004), it was also demonstrated that deviations from this empirical generalisation exist. Kahn *et al.* (1988) were the first to demonstrate that some brands exhibit what were then known as niche and change-of-pace behaviours, while Fader and Schmittlein (1993) brought attention to the triple jeopardy phenomenon, an event that occurs when large MS brands have a behavioural loyalty value greater than that predicted by the Dirichlet model. However, beyond these two seminal studies, many other researchers confirmed the existence of these deviations (Scriven and Bound, 2004; Fang *et al.*, 2006; Pare *et al.*, 2006; Pare and Dawes, 2007; Jung *et al.*, 2010), therefore making it necessary to find a more parsimonious and reliable way to measure consumers' loyalty. This led to the development of the φ.

The application of the φ was largely investigated by Jarvis and Goodman (2005) and Jarvis *et al.* (2006, 2007a, b). The former analysed RP data on the Australian wine market over a population of 4,000 wine shoppers in a one-year period in order to gain a better understanding of the degree of loyalty consumers devote to prices, finding that wines below AUS\$7.50 and above AUS\$17.50 stimulate the highest loyalty levels. Jarvis *et al.* (2006) confirmed these results, while also showing that Australian consumers are very loyal to famous brands, small brands face difficulties in building consumer preference. The analysis was then extended to other product attributes, such as red and white grape varieties and red and white wine regions of origin (Jarvis *et al.*, 2007b). For the first time, it was demonstrated that brands in the wine sector stimulate loyalty less than other product characteristics, such as price, grape variety and region of origin. In particular, it was noted that price is the attribute Australian consumers are most loyal to, followed by white grape varieties, red grape varieties, red wine region of origin and white wine region of origin (Jarvis *et al.*, 2007b). The potentialities of the polarization index were also explored outside the Australian boarders. Casini *et al.* (2009) applied the φ on the purchases of wine made by consumers in the three-year period 2003-2005, finding that the format proves to be the attribute that generates the highest loyalty, followed by quality designation and price. More specifically, table wines, wines sold in the basic tier (<€3) and in bag-in-boxes achieve the highest loyalty values, while foreign wines, wines sold at more than €7 and small bottles are those that discourage loyalty the most. Finally, the results obtained by Krystallis and Chrysochou (2009) on Greek wine consumers through a self-reported questionnaire distributed in March 2009 showed that price, quality certification and winemaker's size stimulate loyalty towards white wines more effectively than reds, while the origin and the grape variety do not constitute a particularly important loyalty component.

To sum up, previous research demonstrated the usefulness of the polarization index for the analysis of loyalty, but none of them tested whether these potentialities hold over time.

3. Methodology

The calculation of the first two BPM – the PF and the SCR – is very straightforward, as it lies on the respective definitions of the two BPM. More precisely, they will be obtained by applying the formulas in (1) and (2):

$$PF_i = \frac{\text{Units purchased of product } i}{\text{Number of buyers of product } i} \tag{1}$$

$$SCR_i = \frac{\text{Units purchased of product } i}{\text{Units purchasd in the category by buyers of product } i} \tag{2}$$

It is possible to obtain ρ by applying the formula defined by Rungie and Laurent (2003):

$$\rho_i = MS_i + \varphi - (MS_i \times \varphi) \tag{3}$$

Equation (3) states that the ρ, the probability of choosing alternative i conditional on a previous purchase of the same alternative i, is function of the MS and the φ, but, more importantly, it shows that MS and φ are two independent measures and that the ρ is only a function of the φ, if the effects of the MS are removed (Rungie and Laurent, 2003).

As a more precise approach, the φ is obtained from the application of the beta binomial distribution (BBD) model to the purchases made by consumers of different brands in the same product category and in a defined interval time. More specifically, it is possible to identify as many φ values – also called in literature marginal φ or BBD values – as the number of brands (or levels of the attribute) in a category. These values express the loyalty level of consumers in the marginal choice between each brand (or level of the attribute) and all the other brands (or other levels of the same attribute) in the category. The analysis then focuses on the deviations of the loyalty for each brand (or level of the attribute) from the average or benchmark loyalty level for the brands (or the various levels of the same attribute) in the category.

Before estimating all the marginal φ, the benchmark value – also called category polarization index (φ_c) or DMD value – needs to be calculated. It can be done as follows:

$$\varphi = \frac{1}{1+S} \tag{4}$$

The index ranges from 0 (maximum disloyalty) to 1 (maximum loyalty) (Rungie and Goodhardt, 2004). In order to solve (4), it is necessary to estimate the value of S from the data. This is obtained by fitting the Dirichlet multinomial distribution (DMD) to the choice of all the brands in a given product category (or all the levels in a given product attribute). The DMD is the multinomial version of the BBD. If there are h brands (or h levels of the attribute), then the DMD has h parameters $\alpha_1, \alpha_2, \ldots, \alpha_h$ and $S = \alpha_1 + \alpha_2 + \ldots + \alpha_h$. In order to estimate the values of parameters $\alpha_1, \alpha_2, \ldots, \alpha_h$, the method of marginal moments can be used (Rungie, 2000), as well as the discrete choice models (Guimaraes and Lindrooth, 2005) or, as in this study, the maximum likelihood theory (Rungie, 2003). Once the estimation of parameters $\alpha_1, \alpha_2, \ldots, \alpha_h$ is obtained for the DMD, the notion S is used to indicate the sum of the j-th values of α.

Once S and φ_c are known, each marginal φ value can be derived. The only difference is that, instead of fitting the DMD simultaneously to the set of all brands, the BBD is fitted to the marginal choice between each brand and all others (or the choice between each level and all other levels of the same attribute). Subsequently, it is possible, within each category, to identify the brands with higher or lower loyalty levels compared to a benchmark level by comparing the polarization for each brand (BBD) with the polarization for the category (DMD).

4. Data

The above-mentioned loyalty measures will be applied on the purchases of wine[4] made by a representative sample of the Italian population in the retail sector, as reported by the AC Nielsen Consumer Panel. The purchases were recorded for six years (2003-2008) and, for the purpose of this research, they were split in two three-year periods: 2003-2005 and 2006-2008. The first sub-sample accounts for 5,299 households, while the second comprises 6,394 families. A further sub-sample was extracted from each of the two groups, in order to include only those households with somewhat regular purchase behaviour. The sub-samples include the families who bought wine on more than one occasion in each of the two three-year periods and bought more than ten units of wine in each of the two interval times. This brought the number of families to 3,858 and 4,643 respectively, and, while in 2003-2005, the final sub-sample has purchased 366,413 litres of wine, in the latter period the population bought 411,638 litres. The following classification was adopted in order to analyse the evolution of loyalty levels towards three product attributes: price, format and quality designation. The decision to select product attributes instead of brands was made based on the results shown in the literature review paragraph.

As for prices, the Rabobank classification (Heijbroeck, 2003) was adopted as a benchmark. This splits wines into the categories of basic (≤€3), popular premium (€3- ≤ €5), premium (€5- ≤ €7), super-premium (€7- ≤ €14), ultra-premium (€14- ≤ €150) and icon (>€150). The first three price ranges represent 68.2, 25.2 and 4.5 per cent of the Italian off-trade market for 0.75 litre bottles. The other three were combined, as they account for a total 2.1 per cent of the market (IRI Infoscan, 2007). Hence, the classification adopted the following levels: <€3, €3- ≤ €5, €5- ≤ €7, and >€7.

For quality designations, an adapted version of the Italian quality classification scheme (Law n. 164/1992) was used, in order to include foreign wines. The Italian regulatory framework gives wineries three main strategic options,

Journal of Product & Brand Management
Volume 20 · Number 2 · 2011 · 111–120

which vary in relation to the level of control on making the wine and the associated costs of meeting the required production standards. At the bottom of the so called "quality pyramid" one finds table wines. These wines must be produced with authorised grapes, but, apart from this, do not have to respect any particular code of production. The latest figures say that these wines represent 37 per cent of the total Italian production (ISTAT, 2009). Above them one finds geographic indication (GI) wines. They must be produced with a minimum of 85 per cent of grapes grown in the area they report on the label. They also need to respect a more restrictive code of productions than table wines, which tends to increase production costs. GI wines represent a 30 per cent of the Italian wine supply (ISTAT, 2009). Finally, one finds *Denominazione di Origine Controllata* (DOC) and *Denominazione di Origine Controllata e Garantita* (DOCG) wines at the top of the pyramid. Quality wines produced in a well-defined area belong to this group. In order to make them, producers have to follow a very strict code of production, differing from one DOC-DOCG designation to the next, which sets a series of rules relative to the production process and the organoleptic and physico-chemical characteristics of the wines. They represent a 33 per cent of the Italian wine production (ISTAT, 2009). To these three quality designations, it is important to add foreign wines. These wines are produced according to the specific rules set by each country, of course, but they are now often present on Italian retail shelves, thus representing another possible option for consumers' choices. Hence, with regards to quality designations, wines were classified in: foreign wine, GI, DOC/DOCG, and table wine.

Regarding formats, wines in 0.75 litre bottles were grouped together. Another cluster was created to represent wines that are generally sold in bottles of 0.2 litres up to 0.5 litres. The other two groups were organised, in order to account for one-litre carton wines, a format largely used in Italy, and larger formats, including 3 litre bag-in-box wines. Hence, the four groups were: <0.75 litres, $=0.75$ litres, >0.75 litres and ≤ 1.5 litres, and >1.5 litres.

5. Results

Table I shows the trends of the three BPM – PF, SCR and ρ – from the three-year period 2003-2005 to 2006-2008 for all the attributes and levels considered in this study with respect to the MS. As one can observe, these trends are all consistent, in the sense that when MS grows from one interval time to the next, the other BPM increase as well, and vice versa. This could lead researchers to think that patterns are clear and there is no need to take into account other loyalty measures. However, if one examines the ρ relative to the <0.75 litre format, foreign wines designation of origin and $>$€7 price, the trends are opposite. Said differently, while MS increase, ρ decreases. However, the difference in MS between the first and the second three-year periods are very small (max $+1.0$ per cent), while the changes in the ρ are greater (more than -3.0 per cent), meaning that these changes are all due to changes in the φ. Therefore, it appears interesting to observe the trends of the φ with more detail.

The observation of the DMD and BBD values both at a category and at a marginal level shows a more complex situation than the BPM would lead researchers to believe exists. At a DMD level, Table II shows that in the three-year period 2006-2008, a significant decrease in the loyalty Italian consumers devoted to the wine attributes under analysis occurred in respect to the former interval time. In particular, the highest loss characterises prices, as the φ_c relative to 2006-2008 registers a decrease of 12 per cent, followed by formats (-8 per cent) and by quality designations (-5 per cent). Nevertheless, the format in which the wine is purchased continues to represent the attribute to which Italian consumers devote the highest loyalty (0.49 in the three-year period 2003-2005 and 0.45 in 2006-2008). This is followed by the quality designation that appears on the label (0.37 and 0.35 respectively) and by price tiers (0.33 and 0.29).

More specifically, the analysis of marginal φ values reveals that the worsening of formats' performances was mainly caused by the >1.5 litre group. In the last three years, this format moved from showing the highest BBD value among all the attributes and levels studied in this research (0.58) to a marginal φ (0.37) similar to that of the <0.75 litre format (0.36). However, while this latter group, which mainly includes 0.375 litre bottles, remained almost stable between 2003-2005 and 2006-2008 both in terms of MS (1.8 and 2.8 per cent) and BBD value (0.36 for both periods), the >1.5 litre tier lost a considerable portion of the market (20.4 and 4.0 per cent). The other two groups, 0.75 litres and >0.75 litres and ≤ 1.5 litres, moved from owning 75 per cent of the market to more than 90 per cent, but they inverted their positions in the ranking, with the former becoming the most purchased format (from 30.5 to 52.7 per cent), while the latter lost 7 per cent circa of MS (from 47.3 to 40.5 per cent). In terms of loyalty, while the 0.75 litre format has been continuing to show excess loyalty, that is a marginal φ higher than the category φ_c, it is interesting to note that the tier >0.75 litres and ≤ 1.5 litres, although stable across the two periods (0.47 for both interval times), reached the same condition only in the last three years.

The results concerning quality designations show that the situation slightly worsened, especially due to the loss of loyalty towards table wines (from 0.46 to 0.45) and DOC-DOCG wines (from 0.40 to 0.37), not compensated by a growth towards GI wines (from 0.24 to 0.25) and foreign wines (from 0.09 to 0.12). Despite this, table wines and DOC-DOCG wines still show excess loyalty, while the other two segments suffer from low loyalty levels. On the contrary, one notes a radical change in terms of MS. Table wines approximately lost seven percentage points (from 53.1 to 38.4 per cent), a share partly absorbed by DOC-DOCG wines (from 30.6 to 37.5 per cent) and GI wines (from 15.7 to 23.1 per cent). Finally, foreign wines remained stable across the two three-year periods (from 0.6 to 1.0 per cent).

Wines sold in the basic tier continue to represent the favourite purchasing group, both in terms of MS and loyalty, but they lost a significant share on both sides. This price range shrank the total purchases by 8 per cent and the BBD value passed from 0.45 to 0.37. The premium (€5-€7) category did not perform well over time. The loss appears to be more limited regarding MS (from 6.1 to 4.1 per cent), but the same cannot be said in respect to loyalty, as the respective marginal φ moved from a value of 0.25 to 0.19. A contrary situation is faced by the popular premium (€3-€5) tier. This increased by almost 9 per cent in terms of MS (from 13.3 per cent in 2003-2005 to 22.7 per cent in 2006-2008), although loyalty to this category remained almost stable (0.29 and 0.30 respectively). Finally, $>$€7 wines remained steady either in terms of MS (1.6 and 2.0 per cent) and loyalty (0.19 and 0.18) over the two periods.

Journal of Product & Brand Management

Volume 20 · Number 2 · 2011 · 111–120

Table I Changes in MS and other BPM from the period 2003-2005 to 2006-2008

Attribute	MS (%) 03-05	MS (%) 06-08	Trend 03/05-06/08	Purchase frequency (litres) 03-05	Purchase frequency (litres) 06-08	Trend 03/05-06/08	SCR (%) 03-05	SCR (%) 06-08	Trend 03/05-06/08	Repeat rate (%) 03-05	Repeat rate (%) 06-08	Trend 03/05-06/08
Format												
<0.75 litres	1.8	2.8	↑	7.8	9.3	↑	8.3	10.2	↑	49.9	46.5	↓
= 0.75 litres	30.5	52.7	↑	33.8	53.2	↑	35.7	58.8	↑	64.5	74.0	↑
>0.75 litres and ≤1.5 litres	47.3	40.5	↓	52.4	43.0	↓	55.3	47.5	↓	73.1	67.3	↓
>1.5 litres	20.4	4.0	↓	55.3	14.1	↓	58.4	15.6	↓	59.4	47.2	↓
Quality designation												
Table wine	53.1	38.4	↓	55.4	39.3	↓	58.3	43.4	↓	70.4	59.9	↓
GI	15.7	23.1	↑	20.1	25.6	↑	21.2	28.3	↑	46.9	50.0	↑
DOC/DOCG	30.6	37.5	↑	35.2	40.1	↑	37.1	44.4	↑	56.3	59.4	↑
Foreign wine	0.6	1.0	↑	3.8	5.5	↑	4.0	6.1	↑	37.4	35.7	↓
Price												
≤€3	79.1	71.3	↓	77.0	65.1	↓	81.1	72.1	↓	85.9	79.6	↓
>€3 and ≤€5	13.3	22.7	↑	18.1	27.0	↑	19.1	29.9	↑	41.6	45.1	↑
>€5 and ≤€7	6.1	4.1	↓	11.3	9.0	↓	11.8	9.9	↓	36.8	31.9	↓
>€7	1.6	2.0	↑	5.9	7.0	↑	6.2	7.7	↑	33.8	30.4	↓

Table II Attribute levels loyalty and MS 2003-2005 and 2006-2008

Attribute	MS 03-05	MS 06-08	Trend 03/05-06/08	DMD (φc) 03-05	DMD (φc) 06-08	Trend 03/05-06/08	BBD (φ) 03-05	BBD (φ) 06-08	Trend 03/05-06/08
Format									
<0.75 litres	1.8	2.8	↑	0.49	0.45	↓	0.36	0.36	–
= 0.75 litres	30.5	52.7	↑	0.49	0.45	↓	0.50	0.48	↓
>0.75 litres and ≤1.5 litres	47.3	40.5	↓	0.49	0.45	↓	0.47	0.47	–
>1.5 litres	20.4	4.0	↓	0.49	0.45	↓	0.58	0.37	↓
Quality designation									
Table wine	53.1	38.4	↓	0.37	0.35	↓	0.46	0.45	↓
GI	15.7	23.1	↑	0.37	0.35	↓	0.24	0.25	↑
DOC/DOCG	30.6	37.5	↑	0.37	0.35	↓	0.40	0.37	↓
Foreign wine	0.6	1.0	↑	0.37	0.35	↓	0.09	0.12	↑
Price									
≤€3	79.1	71.3	↓	0.33	0.29	↓	0.45	0.37	↓
>€3 and ≤€5	13.3	22.7	↑	0.33	0.29	↓	0.29	0.30	↑
>€5 and ≤€7	6.1	4.1	↓	0.33	0.29	↓	0.25	0.19	↓
>€7	1.6	2.0	↑	0.33	0.29	↓	0.19	0.18	↓

As evidenced by Tables I and II, if marketing managers based their decisions exclusively on an examination of BPM, they risk deriving inaccurate conclusions regarding loyalty. Conversely, the φ offers analysts a different perspective on how loyalty evolves over time, which, given its independent nature with respect to MS, appears to be more valid. Next, the implications of this research to marketing managers are discussed.

6. Discussion

By looking at formats, it seems that the very negative performances of the >1.5 litre format, in terms of loyalty and MS, and by 1 litre cartons, in terms of MS, caused the decline of table wines market, rather than the opposite. Although loyalty to formats decreased in the last three years, this attribute is still able to generate a loyalty level appreciably superior to that of quality designations. It is believed that this situation was caused by two main factors. First, the Italian socio-demographic structure has been rapidly evolving. The number of one- and two-member households constitute more than a half of the Italian population. In particular, singles increased from 500,000 in 1995-1996 to 5,100,000 in 2005-2006 and separations/divorces have increased in the last few years (ISTAT, 2007). In this context, the second mistake was that the Italian wine supply chain was not able to convince consumers that if "[you] live alone, and tend to drink a glass or two of wine during dinner once or twice a week, [it is] not quite enough to go through a bottle of wine before it starts to turn over. What [one can] really like about [bag-in-boxes] is their claim that it will last several weeks at room temperature" (Santini *et al.*, 2007, p. 222).

Moreover, the degree of homogeneity in wine demand evolved differently when observed from the viewpoint of loyalty towards formats and that of quality designations. This trend made relatively more loyal either those consumers who purchased 1 litre cartons or 0.75 litre bottles during the three-year period 2003-2005. The BBD values shown by these two attribute levels in 2006-2008 are higher than the DMD value of the respective three-year period, more than the marginal φ registered in 2003-2005 in respect to the corresponding category φ_c. More specifically, data evidence that the Italian wine market moved from having three clear segments of the population in 2003-2005 in respect to the format chosen, to only two groups, hence making >1.5 litre and <0.75 litre tiers occasional purchasing formats. Conversely, the demand of quality designations widened in terms of MS. Table wines, DOC-DOCG and GI wines have more balanced MS than they had in 2003-2005.

Moreover, by looking at MS quotas of each of the four formats and quality designations levels, one notes an almost specular relationship between 1 litre cartons and table wines (40.5 and 38.4 per cent respectively), and between 0.75 litre and DOC-DOCG-GI wines (52.7 and 60.6 per cent respectively). This fact becomes even more prominent under the new EU regulatory framework. Within the new common market organisation (CMO) rules, in fact, a parcel of land cannot host more than one designation, either GI, DOC or DOCG. Therefore, despite the fact that the consumption of quality wines is increasing, all producers, especially smaller ones, of these wines have to think carefully about what they want to produce in the medium- to long-term. It will no longer be possible to move from one designation to another in case the quality of a certain GI, DOC or DOCG does not meet minimum quality standards or the demand is inferior to the supply in a given year. It will only be possible to declass a GI, DOC or DOCG wine to table wine, an operation that has significant economic and financial implications beyond the loss in terms of image and brand perception that such a choice could determine.

The evolution of loyalty to prices also offers an interesting topic for discussion. First, the 2003-2005 results showed that the popular premium and the premium price tier have similar loyalty values, although lower to that of the category. However, when grouped together, their BBD value is higher than those of the two taken separately. This suggests that Italian consumers tended to shift from one price level to the other, especially when the price is closed to the delimiting value (€5). Moreover, in case some events occur, (for example, price promotions, higher distributing power of producers, and better placement on the shelf) consumers could be pushed to one tier or the other. Conversely, by examining the 2006-2008 results, it appears probable that part of the MS and definitely most of the loyalty stimulated by the premium price category was cannibalised by the popular premium one. The exaggerate use of promotions made consumers understand that a certain number of premium price brands are always discounted by, for example, 20 per cent, making them popular premium brands. This means that if a certain category of wines, either DOC-DOCG or GI, able to satisfy consumers more or less to the same extent can always be found at a lower price, why should consumers spend more to buy them? Consequently, while waiting the best-buy-of-the-week, consumers widened the number of brands they purchased in the latter three years, premium wines reduced their presence in the market, and popular premium wines increased both their loyalty and MS. Second, the Italian retail wine sector traded up in the latter three years, moving from a consumption mainly based on lower price wines to popular premium ones, although table wines sold at less than €3 in one-litre cartons or bag-in-boxes still remain the current market leaders. Secondly, the position of <0.75 litre format and $>$€7 wines did not change much in the last three years. Not many years ago, Anthony Rose stated that "as customers spend more on wines, supermarkets need fine wines to raise the image of their ranges" (Rose, 2001). A few years later, his words seem to be still relevant. Finally, super-premium, ultra-premium, icon and foreign wines are present on supermarkets shelves to improve their image, but they continue to represent an exception for a vast majority of consumers, who generally buy products they are more familiar with or cost less. However, this situation should not worry market analysts. The vast majority of producers of luxury wine brands prefer to target specialised shops and the on-trade sector, as these distribution channels may be more profitable for them.

7. Managerial implications

The results of this study lead to three main managerial considerations for wine producers, marketers and the industry in Italy.

First, producers must decide whether to follow a more basic or a more premium strategy to production. The former is characterised by wines sold below €3, in 0.75 litre bottles or 1 litre cartons, without any designation of origin, neither GI, DOC nor DOCG. This strategy could be better undertaken by major producers, who are able to provide retailers with a huge supply of wine. Moreover, this approach better enables promotions like discounts, buy-one-get-one-free, etc. These products, in fact, are already positioned in a high loyalty price tier, hence the demand associated with them will remain inelastic. By contrast, smaller wineries could focus more on the latter premium strategy. This means choosing a GI-DOC-DOCG wine with a 0.75 litre bottle and priced between €3 and €5.

Furthermore, with the premium strategy there must be a decision between conveniently continuing to produce a DOC-DOCG wine or moving to the expanding market for GI wines. A suggested pattern to follow could be to conduct an economic/financial analysis able to provide an indication of the strength of the quality designation one has the rights to produce. In case the results showed that the designation is strong (which is likely to be probable for several DOC-DOCG such as the different typologies of *Chianti*, the *Brunello di Montalcino* or the *Amarone della Valpolicella*), then it could be strategically correct to continue with it. Conversely, if the value of a designation is weak (which is often the case for several of those born in the last ten or fifteen years), the advice is to change to a GI. Conversely, a GI designation is able to offer consumers a clear and evident link with the territory, the grape varieties used and the possible pairings with a dish, all elements found to be important for a choice of a wine. Simultaneously, GI wines will sensibly reduce production costs with respect to a DOC-DOCG, as the rules of production are less restrictive. Moreover, these rules are not only less restrictive, but they also offer more flexibility, thus giving producers the possibility to adapt a wine according to the ever-changing tastes of the population.

The choice between GI and DOCG for individual premium producers leads to the second managerial implication. A further move to GI production will have impact on the whole country and industry. It will lead to a rationalisation and a reduction in the number of DOCs and DOCGs designations currently active. It would be suicide for the Italian wine sector to maintain, or, even worse, to increase the current number of the 364 DOC and DOCG designations now present in the market. Conversely, a reduction will make them even more exclusive, augmenting the value perceived by consumers.

The third managerial implication refers to the use of the polarization, φ. Managers interested in trends for the loyalty consumers attach to a specific brand or product attribute, can use the classical loyalty BPM, such as the PF, the SCR and the repeat purchase rate. However, before any consideration is drawn from these BPM, managers should also carefully observe the results of the φ. Where the different measures of loyalty are not consistent, it is better to use the polarization index, due to its independent nature with regards to market share.

8. Conclusions

This study showed that although BPM are certainly a fundamental source of information about the degree of loyalty consumers attach to brands and products, their strong relationship with MS risk providing results that do not always reflect real loyalty trends. Conversely, the φ proved valid in analysing changes in loyalty evolves over time. Therefore, marketing managers should always compare the results obtained with the BPM and with the φ before drawing conclusions on the actual loyalty trends of their products and brands.

More specifically, from a managerial perspective the analysis of loyalty dynamics through the φ relative to the two three-year periods 2003-2005 and 2006-2008 registers a significant decrease, especially in respect to the prices at which wines are purchased. In particular, formats continue to represent the attribute towards which Italian consumers devote the highest loyalty, followed by the quality designation that appears on the label and by price tiers. The bag-in-box format almost disappeared from the market, the purchases of 0.75 litre bottles increased in terms of loyalty and MS, while 1 litre cartons only lost a few percentage points of share. The consumption of table wines decreased, substituted by DOC-DOCG and GI wines. Concerning prices, wines sold in the basic tier continue to represent the most favourite purchasing group, in terms of both MS and loyalty, but they lost a significant share on both sides. The premium category did not perform well over time, especially due to a loss in terms of loyalty, while the popular premium tier benefited from an increase in sales, despite remaining stable on a loyalty dimension. Finally, >€7 wines did not vary much over the two three-year period, still playing a marginal role in the Italian retail wine sector.

Future research will continue to focus on the way in which loyalty evolves over time, by testing this methodology on other food and beverage categories. Moreover, different techniques will be applied to these panel data, in order to demonstrate the consistency of the results obtained so far. Finally, research will be conducted on the way in which product attributes interact or correlate, thus stimulating higher or lower MSs and behavioural loyalty values.

Notes

1 The PF is defined as the number of times a shopper made a purchase of a brand/product within a specified period, and it is often expressed as an average purchasing rate across all those who have bought the brand at least once within the specified period (Uncles *et al.*, 2010).
2 The SCR represents the market share of a brand among all the consumers who purchased the brand at least once during a specific period (Jung *et al.*, 2010).
3 ρ is defined as "the probability of choosing alternative i conditional on a previous purchase of the same alternative i" (Jarvis *et al.*, 2007a, p. 495).
4 They refer to all the purchases of Italian and foreign wines with an EAN code made by the AC Nielsen Consumer Panel. These data do not include non-EAN wines and the following categories: champagne, marsala, sherry, port, grape must, wine-based aromatised beverages, sangria, aromatised wines, natural sparkling wines, fortified wines, and the spumante category. Hence, the data represents for AC Nielsen, the 71 per cent in volume and the 78 per cent in value of the wine purchased in Italy (average 2005-2009) in the retail channel.

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Further reading

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Executive summary and implications for managers and executives

This summary has been provided to allow managers and executives a rapid appreciation of the content of this article. Those with a particular interest in the topic covered may then read the article in toto *to take advantage of the more comprehensive description of the research undertaken and its results to get the full benefits of the material present.*

Both marketers and business organizations attach great importance to consumer loyalty. A plethora of studies have indicated a range of benefits firms can secure through retaining their customers. For instance, loyal patrons tend to spend higher amounts and make more frequent purchases. It is more expensive to attract new customers than keep existing ones who often act as brand advocates through positive word-of-mouth recommendations. Loyalty patrons are also less deal prone, while showing a willingness to overlook the occasional product failure. In addition, firms that understand the preferences of regular customers are able to improve their marketing efficiency as a result.

Measuring loyalty is therefore crucial and many marketers use various brand performance measures (BPM) for this purpose. The measures can provide organizations with insight into the behavioural loyalty of consumers reflected by purchase frequency (PF), share of category requirements (SCR) and repeat purchase behaviour (RP). Some studies have shown that products or brands tend to rate higher in these factors when their market share (MS) is greater. This correlation is not irrefutable though, as plenty researchers have found.

Loyalty evolves over time and a common criticism of BPM is that the method is unable to accurately reveal a developing trend. And while BPM are able to provide some information relevant to consumer loyalty, scholars have noted certain limitations. Collecting data can be expensive and laborious, information often overlaps and interpreting the results is difficult. Overall, a genuine possibility exists that using BPM alone to estimate loyalty changes may generate unreliable indications.

The need for a more appropriate measure of loyalty has inspired the development of the polarization index (PI). This measure functions independently to MS and has been

Journal of Product & Brand Management
Volume 20 · Number 2 · 2011 · 111–120

successfully deployed in studies of consumer behaviour regarding wine in Australia, Greece and Italy. This research focused on periods of either one or three years and some significant findings were produced. One study into Australian consumers indicated their loyalty to well-known brands, while later research found that loyalty was greater to certain product attributes such as price, grape variety and region of origin in that order. An Italian study using the PI discovered format to have the greatest impact on loyalty, with quality designation and price close behind. Price, quality certification and winemaker size were found most influential on the loyalty of Greek wine consumers. The impact was greater on white than red wines and the study also showed that grape origin and variety were largely insignificant.

In the present work, Corsi *et al.* compare findings produced by BPM and PI concerning wine purchases made by Italian consumers during the separate periods 2003-2005 and 2006-2008 in order to measure longer-term trends. The retail sector in Italy was perceived as a relevant context for the research, not least because of the challenging climate as indicated by falling levels of wine consumption within the country. The authors provide various equations and formulae to indicate how PI is calculated and the measure is applied to the data obtained.

A consumer panel was used to acquire a representative sample of Italian wine consumers for the two three-year periods under analysis. Comprehensive samples were generated for each period as were sub-samples of respondents who purchased wine on a more regular basis. Following earlier results indicated in the literature, the decision was taken to investigate loyalty towards the price, format and quality designation characteristics rather than to specific brands.

Regarding price, an established method was used to classify wines into four distinct categories ranging from below €3 at one end to above €7 at the other. The Italian quality classification scheme helped form categories that included table wines at the foot of the quality pyramid and *Denominazione di Origine Controllata* (DOC) and *Denominazione di Origine Controllata e Garantita* (DOCG) at the top. The other categories created were foreign wines and geographic indication (GI) wines. The authors point out the manufacturing obligations of each wine quality and its percentage of total Italian wine production. With format, size was used to identify four distinct groups.

Analysis based on BPM indicates strong correlation between the measures and MS in terms of both increase and decrease. However, a different picture often emerges when PI enters the equation. One example is the contrasting trend regarding wines priced above €7, foreign wines and the below 0.75 litre format. It is additionally evident that MS changes in the two periods are minimal, while PI differences are considerably greater.

PI analysis revealed a decrease in loyalty to formats over the second period, although the level of loyalty was still stronger than that towards quality designation or price. In part, this is attributed to the changing socio-demographic profile of the Italian population that has seen an increase in single-person households due to higher divorce rates. One outcome of this has been a considerably lower demand for the bag-in-box format.

Other indications include a reduction in the number of segments in respect of the format chosen and almost "specular relationship" between table wines and 1 litre cartons and between DOC, DOCG and GI wines with the 0.75 litre format in terms of MS quotas. Changing in loyalty towards prices is also seen as significant. The authors conjecture that Italian consumers may switch between price levels and that this behaviour can be influenced by promotional activities. Promotions might also impact on the sales of higher quality wines if consumers believe that they can purchase similar quality products at a lower price. Another change revealed in the study was a shift in consumption from lower price wines to those labelled as "popular premium". It is, nevertheless, pointed out that table wines retailing at under €3 retained their market leading position.

Based on the study indications, Corsi *et al.* claim that wine producers need to decide between "more basic or more premium" product strategies. The basic approach could be more suitable to larger producers able to supply massive quantities of the product. It is further suggested that scope exists with this approach to offer discounts and other incentives without negatively affecting the already high demand and loyalty levels. Smaller producers may likewise be better served by the more premium strategy because of the lower demand for these products.

Producers adopting a more premium strategy need to decide whether to stick with DOC-DOCG wines or shift to the GI category that has become an expanding market. The authors suggest conducting a financial analysis to determine the appropriate quality level. Shifting to GI helps to lower costs as fewer restrictions surround production. That there is greater flexibility also gives producers the option of adapting its wines to account for changes in tastes. Moving to GI and accordingly reducing the number of DOC and DOCG producers can help increase both the exclusivity and perceived value of these higher quality wines.

Marketers are urged to compare the indications of BPM with PI to obtain a more informative picture of loyalty towards their products and brands. Using the methodology in relation to other food and beverage categories is one option for further research, which might also investigate the interaction of product attributes and the resulting effect on MS.

(A précis of the article "Is the polarization index a valid measure of loyalty for evaluating changes over time?". Supplied by Marketing Consultants for Emerald.)